Diagnose the sjSDM Setup

Description

Comprehensive non-aborting diagnosis of the sjSDM installation, including the R console, Python, PyTorch, CUDA support, and sjSDM functionality.

Usage

```
diagnose_sjsdm_setup(run_test_model = base::interactive())
```

Arguments

`run_test_model`	Logical. Should a test model be fitted to diagnose full functionality? Defaults to 'interactive()'.

Details

This function performs the following checks: 1. R console: RStudio, radian, or plain R ('Rscript' or the R console) 2. Python version and location, with reticulate's message when no usable Python is found 3. PyTorch installation and version 4. CUDA/GPU availability 5. sjSDM package installation 6. sjSDM Python dependencies 7. Test model fitting (optional)

R running inside radian shares its process with radian's Python, so reticulate must use that Python and the project 'RETICULATE_PYTHON' is replaced by it. radian is therefore judged by whether its Python can import PyTorch, not by whether a radian executable is on 'PATH'. Outside radian, reticulate follows 'RETICULATE_PYTHON'.

All checks print their status with [OK], [WARN], or [FAIL]. Failures are represented in the returned list rather than raised as errors. Fixes point to 'Documentation/Materials/sjSDM_installation_guide.md'.

Value

Invisible named list of logical flags: 'radian_ok', 'python_ok', 'pytorch_ok', 'cuda_available', 'sjsdm_ok', and 'test_model_ok'. ‘radian_ok' is 'FALSE' only when R runs inside radian and radian’s own Python cannot import PyTorch; it is 'TRUE' in RStudio and plain R. Called primarily for its printed report.

See Also

`py_config` `sjSDM`